# Stochastic Modelling of Life Expectancy and Longevity Risk Quantification in a Life Annuity Portfolio

Comparative Analysis of Actuarial Models on French Data 1950–2024: Lee-Carter, CBD, Renshaw-Haberman and Bayesian Extension

**Institut National de Statistique et d'Économie Appliquée (INSEA)**
Department of Statistics, Biostatistics, Demography & Big Data
Rabat, Morocco

**Data:** Human Mortality Database — France Males & Females 1950–2024
**Keywords:** Lee-Carter · CBD · Renshaw-Haberman · Kalman Filter · Longevity risk · Life annuity · Rolling backtest · Solvency II

2024

**Abstract**

This paper presents a systematic comparison of four stochastic mortality modelling approaches on French male and female data over the period 1950–2024, with application to longevity risk quantification in a life annuity portfolio.

We calibrate and validate the Lee-Carter (1992), Cairns-Blake-Dowd (CBD, 2006) and Renshaw-Haberman (2006) models, supplemented by a Bayesian extension of Lee-Carter based on a Kalman filter and a two-factor formulation (LC2) motivated by the bimodal structure of male mortality. Validation is conducted via a rolling one-step-ahead backtest over 2001–2020, using RMSE, bias and 95% confidence interval coverage as performance metrics.

The CBD model achieves the best predictive performance, with RMSE of 0.188 years (females) and 0.228 years (males), and coverage of 91.3% and 95.7% respectively. The Renshaw-Haberman model minimises absolute bias (+0.052 years for females), confirming a statistically significant cohort effect in French mortality. The LC2 extension improves male coverage by +21.8 percentage points over the single-factor Lee-Carter model.

On the application side, we build a life annuity pricing function $\ddot{a}_x$ from projected mortality tables and quantify longevity risk for a portfolio of 100 contracts representing 14.37 M€ in committed capital. Results indicate that a stochastic provision (P99.5) of €71,298 (0.5% of capital) is required to cover parametric uncertainty, while the Solvency II regulatory stress ($-20\%$ on $m_x$) generates a surplus engagement of €524,015 (3.6% of capital), bringing the total recommended provision to €595,313 (4.1% of committed capital).

**Keywords:** Lee-Carter; CBD; Renshaw-Haberman; Kalman filter; life expectancy; rolling backtest; life annuity; longevity risk; Solvency II; Human Mortality Database.

**JEL Classification:** G22, J11, C53, C11.

# Contents

**1 Introduction** **3**
1.1 Background and motivation . . . 3
1.2 Research questions . . . 3
1.3 Main contributions . . . 3

**2 Data** **4**
2.1 Source and structure . . . 4
2.2 Methodological choices . . . 4

**3 Methodology** **5**
3.1 Life table . . . 5
3.2 Lee-Carter model (LC1) . . . 5
3.3 Two-factor Lee-Carter model (LC2) . . . 5
3.4 Cairns-Blake-Dowd model (CBD) . . . 6
3.5 Renshaw-Haberman model (RH) . . . 6
3.6 Bayesian extension — Kalman Filter . . . 6
3.7 Bootstrap confidence intervals . . . 7
3.8 Validation protocol — Rolling backtest . . . 7

**4 Results** **7**
4.1 Lee-Carter parameters . . . 7
4.2 Residual analysis . . . 7
4.3 CBD parameters . . . 8
4.4 Backtest results . . . 8
4.5 Projected life expectancy in 2025 . . . 9

**5 Actuarial Application** **9**
5.1 Life annuity pricing . . . 9
5.2 Portfolio longevity risk quantification . . . 10

**6 Discussion** **11**
6.1 Comparison with the literature . . . 12
6.2 Limitations . . . 12
6.3 Future work . . . 12

**7 Conclusion** **13**

# 1. Introduction

Life expectancy is the central metric in life actuarial science. Any insurance company offering life annuities, life insurance or retirement products must project future mortality to price products correctly and build adequate reserves. Underestimating mortality improvement results in longer-than-expected annuity payments, exposing the insurer to *longevity risk* (Blake et al., 2006).

## 1.1. Background and motivation

France has one of the highest life expectancies in Europe, with 87.2 years for females and 80.1 years for males in 2022. Mortality improvement has been steady since 1950, punctuated by identifiable exogenous shocks: the AIDS epidemic (1985–1995), the 2003 heatwave and the COVID-19 pandemic (2020). These shocks affect the sexes asymmetrically: male mortality exhibits a bimodal structure, with shocks concentrated at ages 20–45 (external causes) and a regular trend at ages 60–90 (chronic diseases), which single-factor models struggle to capture.

The Human Mortality Database (HMD) provides exceptional quality data since 1816, enabling robust calibration of stochastic models. We restrict the analysis to French data over 1950–2024, distinguishing males from females, to avoid pre-1950 regime biases.

## 1.2. Research questions

**Q1.** Which stochastic model minimises life expectancy forecasting error over 2001–2020, for both sexes?

**Q2.** Does adding a second factor to the Lee-Carter model correct the insufficient confidence interval coverage for males?

**Q3.** What is the actuarially fair premium for a life annuity in 2025, and to what extent does it depend on the model chosen?

**Q4.** What longevity provision must an insurer hold to satisfy Solvency II requirements?

## 1.3. Main contributions

**C1.** A systematic comparison of five models (LC1, LC2, CBD, RH and Bayesian Kalman) on an identical validation protocol applied to recent French H+F data

(up to 2024).

**C2.** An LC2 formulation motivated by residual analysis of Lee-Carter for males, achieving a 19.2% reduction in residual standard deviation.

**C3.** A longevity risk quantification distinguishing stochastic uncertainty (bootstrap P99.5 provision) from deterministic regulatory stress (Solvency II shock), with operational recommendations.

## 2. Data

### 2.1. Source and structure

Data come from the Human Mortality Database (HMD, 2024), which compiles period mortality tables by age and sex from French civil registration records and national censuses. Table 1 summarises the data structure.

Table 1: HMD data structure — France 1950–2024

| Variable | Range | Definition |
|---|---|---|
| $m_{x,t}$ | $[10^{-5}, 0.40]$ | Central mortality rate: $D_{x,t}/E_{x,t}$ |
| $D_{x,t}$ | Non-negative integers | Observed deaths at age $x$ in year $t$ |
| $E_{x,t}$ | Positive reals | Exposure in person-years |
| Age ($x$) | 0–90 | Exact age (annual groups) |
| Year ($t$) | 1950–2024 | Calendar year |

### 2.2. Methodological choices

**Time restriction.** The calibration period (1950–2000) is chosen to avoid pre-1950 regime biases. The validation period (2001–2020) covers 20 years including major exogenous shocks (COVID-19, 2003 heatwave).

**Age restriction.** We restrict to ages 0–90. Beyond 90, exposures are insufficient. CBD is applied on ages 50–85, the validity zone of the logit linearity assumption.

**Descriptive statistics.** Female life expectancy rose from 69.4 years in 1950 to 85.7 years in 2022 (+16.3 years). Male life expectancy rose from 63.4 to 80.1 years (+16.7 years). The gender gap peaked at 8.5 years around 1980 (AIDS epidemic, road accidents) before converging to 7.0 years in 2022.

# 3. Methodology

## 3.1. Life table

The life table transforms central mortality rates $m_{x,t}$ into a cohort of $l_0 = 100,000$ births:

$$q_x = \frac{m_x}{1 + (1 - a_x)\, m_x}, \quad l_{x+1} = l_x \cdot (1 - q_x), \quad L_x = l_x - (1 - a_x)\, d_x \tag{1}$$

$$T_x = \sum_{j=x}^{\omega} L_j, \quad e_x = \frac{T_x}{l_x} \tag{2}$$

where $a_x = 0.5$ (UDD assumption) except at age 0 (Coale-Demeny correction).

## 3.2. Lee-Carter model (LC1)

Lee & Carter (1992) propose decomposing the log mortality rate as:

$$\log m_{x,t} = a_x + b_x \cdot k_t + \varepsilon_{x,t}, \quad \varepsilon_{x,t} \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{3}$$

where $a_x$ is the average age profile, $b_x$ the age-specific sensitivity to the time index $k_t$.

**SVD calibration.** Parameters are estimated via singular value decomposition (SVD) of the centred matrix $\log m_{x,t} - a_x$. Identifiability constraints $\sum_x b_x = 1$ and $\sum_t k_t = 0$ ensure uniqueness.

**$k_t$ projection.** $k_t$ is projected via ARIMA, selected automatically by AIC criterion. Over the calibration period, $k_t$ declines from $+56.8$ (1950) to $-49.7$ (2000).

## 3.3. Two-factor Lee-Carter model (LC2)

Residual analysis of LC1 reveals a systematic pattern for males at ages 20–45 (a red band in the heatmap), not captured by the single factor $k_t$. We propose:

$$\log m_{x,t} = a_x + b_x^{(1)} k_t^{(1)} + b_x^{(2)} k_t^{(2)} + \varepsilon_{x,t} \tag{4}$$

Both components are obtained from the first two SVD singular values. The second factor $b_x^{(2)}$ presents a characteristic peak at ages 20–45 for males, corresponding to

the AIDS epidemic and road accidents.

### 3.4. Cairns-Blake-Dowd model (CBD)

Cairns et al. (2006) model the logit of the mortality probability as a linear function of centred age:

$$\operatorname{logit} q_{x,t} = \kappa_1(t) + \kappa_2(t) \cdot (x - \bar{x}), \quad \bar{x} = 67.5 \tag{5}$$

For each year $t$, $\kappa_1(t)$ and $\kappa_2(t)$ are estimated by OLS on 36 data points (ages 50–85). Both factors are then projected independently by ARIMA.

**Age range selection.** The original model uses ages 40–90. Our tests show that beyond age 85, the logit-age relationship is no longer linear (super-exponential mortality), generating a systematic bias of +0.93 years in projected $e_0$. Restricting to 50–85 reduces this bias to +0.13 years.

### 3.5. Renshaw-Haberman model (RH)

Renshaw & Haberman (2006) extend Lee-Carter by adding a cohort term $\gamma_{t-x}$:

$$\log m_{x,t} = a_x + b_x \cdot k_t + \gamma_{t-x} + \varepsilon_{x,t} \tag{6}$$

**Two-step implementation.** Full alternating least squares (ALS) proves numerically unstable (Haberman & Renshaw, 2011). We use a two-step approach: (1) standard SVD Lee-Carter, then (2) $\gamma_c$ estimated as the mean residual for each cohort $c = t - x$ with at least 5 observations. This reduces residual standard deviation from 0.0975 to 0.0748 (−23.3%), confirming a statistically significant cohort effect.

### 3.6. Bayesian extension — Kalman Filter

We reformulate Lee-Carter as a state-space model:

$$k_t = k_{t-1} + \delta + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2_{\text{proc}}) \tag{7}$$

$$k_t^{\text{obs}} = k_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2_{\text{obs}}) \tag{8}$$

where $k_t^{\text{obs}}$ is the noisy observation of $k_t$ obtained by projecting data onto $b_x$. Hyperparameters $\sigma_{\text{proc}} = 1.548$ and $\sigma_{\text{obs}} = 1.353$ are estimated by maximum likelihood (Nelder-Mead). The Kalman gain $K = P_{\text{pred}}/(P_{\text{pred}} + \sigma^2_{\text{obs}})$ weights the update between model prediction and data observation.

### 3.7. Bootstrap confidence intervals

For each model, confidence intervals on $e_0$ are constructed via parametric bootstrap from three uncertainty sources:

1. $k_t$ **forecast uncertainty**: $k_t \sim \mathcal{N}(\hat{k}_t, \hat{\sigma}^2_{\text{ARIMA}})$
2. **Model residual**: $\varepsilon_{x,t} \sim \mathcal{N}(0, \hat{\sigma}^2_\varepsilon)$, where $\hat{\sigma}_\varepsilon = 0.0975$ (females) and 0.0980 (males)
3. $b_x$ **uncertainty**: Gaussian perturbation calibrated on the estimated standard error from residuals

### 3.8. Validation protocol — Rolling backtest

At each test year $t \in [2001, 2020]$, the model is re-estimated on all available history up to $t-1$, predicts $k_t$, reconstructs $m_{x,t}$ and computes $e_0(t)$. Then observed $k_t$ is added to history. Validation metrics:

$$\text{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\hat{e}_0(t) - e_0(t))^2}, \quad \text{Bias} = \frac{1}{T}\sum_{t=1}^{T}(\hat{e}_0(t) - e_0(t)), \quad \text{Coverage} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{1}(e_0(t) \in \text{CI}_{95\%}) \tag{9}$$

## 4. Results

### 4.1. Lee-Carter parameters

Table 2: Lee-Carter calibrated parameters — France 1950–2000

| Parameter | Females | Males |
|---|---|---|
| $a_x$ range | $[-8.47\,;\,-1.53]$ | $[-8.21\,;\,-1.48]$ |
| $b_x$ range | $[0.004\,;\,0.022]$ | $[0.004\,;\,0.020]$ |
| $k_t$ in 1950 | $+56.8$ | $+52.3$ |
| $k_t$ in 2000 | $-49.7$ | $-37.1$ |
| ARIMA drift | $-2.117\,\text{yr}^{-1}$ | $-1.784\,\text{yr}^{-1}$ |
| Residual std | 0.0975 | 0.0980 |

### 4.2. Residual analysis

The Lee-Carter residuals heatmap reveals three systematic patterns:

1. **Red diagonal band (ages 40–70, 1950→1990)**: cohort effect. Generations born 1910–1930 show excess mortality unexplained by the period trend (WWI sequelae, 1918 Spanish flu).
2. **Blue block at ages 60–90 after 1985**: underestimated cardiovascular improvement. Statins, bypass surgery and hypertension treatment massively benefited the elderly from the 1980s onward.
3. **Red block at ages 15–35 for males, 1985–1995**: AIDS epidemic and road accident peak. A structural shock no extrapolative model can anticipate.

The LC2 extension reduces residual standard deviation from 0.0980 to 0.0792 for males (−19.2%), with $b_x^{(2)}$ showing a characteristic peak at ages 20–45.

### 4.3. CBD parameters

Table 3: CBD calibrated parameters — Ages 50–85, France 1950–2000

| **Parameter** | **Females** | | **Males** | |
|---|---|---|---|---|
| | 1950 | 2000 | 1950 | 2000 |
| $\kappa_1(t)$ | −3.60 | −4.47 | −3.00 | −3.66 |
| $\kappa_2(t)$ | 0.1055 | 0.0983 | 0.0905 | 0.0872 |
| Residual std (logit) | 0.1342 | | 0.0832 | |

The decline in $\kappa_2$ for both sexes indicates that the logit-age gradient is flattening over time: the elderly are improving their mortality relatively faster, consistent with the post-1985 cardiovascular revolution.

### 4.4. Backtest results

Table 4 summarises rolling backtest results over 2001–2020.

Table 4: Rolling backtest results 2001–2020 — France Males & Females

| **Model** | **Females** | | | **Males** | | |
|---|---|---|---|---|---|---|
| | RMSE | Bias | Cov. | RMSE | Bias | Cov. |
| Lee-Carter (LC1) | 0.463 | +0.148 | 82.6% | 0.636 | −0.552 | 47.8% |
| Lee-Carter 2F (LC2) | 0.601 | +0.374 | 78.3% | 0.556 | −0.441 | 69.6% |
| CBD | **0.188** | +0.084 | 91.3% | **0.228** | −0.087 | **95.7%** |
| Renshaw-Haberman | 0.384 | **+0.052** | 87.0% | 0.562 | −0.471 | 60.9% |
| LC Bayesian (KF) | 0.482 | +0.139 | **95.7%** | 0.692 | −0.624 | 65.2% |

RMSE and Bias in years. Cov. = 95% CI coverage. **Bold** = best in category.

**CBD dominates on both sexes.** CBD achieves the best RMSE for both females (0.188 years) and males (0.228 years), and the best male coverage (95.7%). Its logit-age formulation is better suited to old ages, where most $e_0$ variability occurs.

**Structural insufficiency of Lee-Carter for males.** The LC1 coverage of 47.8% for males means half of observed years fall outside the 95% CI. This is structural: the single factor $k_t$ cannot simultaneously capture volatile shocks at ages 20–45 and the regular trend at ages 60–90. The LC2 extension improves coverage by +21.8 points (47.8% → 69.6%) without reaching CBD levels.

**Renshaw-Haberman — best bias for females.** The bias of +0.052 years confirms that cohort correction better centres female forecasts. The residual variance reduction (−23.3%) confirms the statistical significance of the French cohort effect.

### 4.5. Projected life expectancy in 2025

Table 5: $e_0$ projected to 2025 by model

| Model | Females (yr) | Males (yr) | Gender gap (yr) | Model risk |
|---|---|---|---|---|
| Lee-Carter | 89.001 | 79.535 | 9.47 | — |
| CBD | 89.073 | 79.145 | 9.93 | — |
| Renshaw-Haberman | 89.497 | 79.619 | 9.88 | — |
| Inter-model spread (F) | 0.496 | | | ±0.25 yr |
| Inter-model spread (M) | | 0.474 | | ±0.24 yr |

The inter-model spread on projected $e_0$ is below 0.5 years for both sexes, reflecting the robustness of projections. This spread is a direct measure of *model risk*, which translates into euros in the next section.

## 5. Actuarial Application

### 5.1. Life annuity pricing

The present value of an immediate life annuity from age $x$ is:

$$\ddot{a}_x = \sum_{k=0}^{\omega} v^k \cdot {}_kp_x, \quad v = \frac{1}{1+i}, \quad {}_kp_x = \frac{l_{x+k}}{l_x} \tag{10}$$

with technical rate $i = 2\%$ and maximum horizon $\omega = 40$ years. The fair annual premium is $P = \text{Capital}/\ddot{a}_x$.

Table 6: Life annuity pricing — Female and Male aged 65, Capital €100,000, $i = 2\%$, 2025

| **Model** | **Female 65** | | **Male 65** | |
|---|---|---|---|---|
| | $\ddot{a}_x$ | Premium (€/yr) | $\ddot{a}_x$ | Premium (€/yr) |
| Lee-Carter | 17.994 | 5,558 | 15.847 | 6,312 |
| CBD | 17.969 | 5,565 | 15.623 | 6,401 |
| Renshaw-Haberman | 18.037 | 5,544 | 15.894 | 6,293 |
| Max spread | 0.068 | 21 | 0.271 | 108 |

Model convergence is remarkable: the maximum spread on the annual premium is €21 for females (±0.4%) and €108 for males (±1.7%).

**Longevity VaR via bootstrap (CBD).** Over 1,000 CBD bootstrap simulations, the $\ddot{a}_x$ distribution yields: $P_5 = 17.364$, $P_{50} = 17.969$, $P_{95} = 18.475$. The prudential premium (P95) is €5,759/year, representing +3.1% above the central premium.

### 5.2. Portfolio longevity risk quantification

**Portfolio description.** We consider a portfolio of 100 immediate life annuity contracts with characteristics summarised in Table 7. Capitals follow a log-normal distribution ($\mu = \ln 120{,}000$, $\sigma = 0.5$), representative of a French supplementary retirement portfolio.

Table 7: Portfolio characteristics

| **Indicator** | **Value** |
|---|---|
| Number of contracts | 100 |
| Total committed capital | €14.37 M |
| Mean capital | €143,653 |
| Female/male ratio | 66% / 34% |
| Mean age | 66.5 years |
| Age range | 57–79 years |
| Annual annuity total (CBD) | €921,735/year |

**Stochastic provision.** The portfolio $\ddot{a}_x$ distribution is obtained from 1,000 bootstrap simulations on CBD parameters. Table 8 presents the longevity risk provisions.

Table 8: Longevity risk provisions — Portfolio of 100 annuities, €14.37 M

| **Scenario** | **Mean $\ddot{a}_x$** | **Provision (€)** | **% Capital** | **Use** |
|---|---|---|---|---|
| Central (P50) | 16.087 | 0 | 0.0% | Pricing |
| Prudent (P75) | 16.116 | 0 | 0.0% | — |
| High (P90) | 16.143 | 18,681 | 0.1% | — |
| Severe (P95) | 16.155 | 29,359 | 0.2% | Recommended |
| Extreme (P99) | 16.192 | 62,441 | 0.4% | — |
| Solvency II (P99.5) | 16.202 | 71,298 | 0.5% | Regulatory |

**Regulatory stress tests.** The Solvency II longevity stress consists of a sudden −20% reduction on all projected $m_{x,t}$. Table 9 presents results for different shock levels.

Table 9: Stress tests — Impact of a sudden mortality improvement

| **Mortality shock** | **Surplus engagement (€)** | **% Capital** | **Solvency II** |
|---|---|---|---|
| −5% | 126,739 | 0.9% | Below threshold |
| −10% | 256,257 | 1.8% | Below threshold |
| −15% | 388,650 | 2.7% | Below threshold |
| −20% (regulatory shock) | 524,015 | 3.6% | **Reference** |
| −25% | 662,454 | 4.6% | Extreme |

**Final recommendation.** The stochastic provision (P99.5 = €71,298, 0.5% of capital) covers parametric model uncertainty. The Solvency II deterministic stress (€524,015, 3.6% of capital) covers the regulatory scenario. An insurer must hold both:

$$\text{Total provision} = \underbrace{71{,}298\,€}_{\text{stochastic}} + \underbrace{524{,}015\,€}_{\text{Solvency II stress}} = \boxed{595{,}313\,€ \approx 4.1\% \text{ of committed capital}} \tag{11}$$

**Model risk.** The Lee-Carter versus CBD spread on portfolio valuation is €62,027 (0.4% of capital). Using a sub-optimal model thus generates a directly quantifiable pricing error.

## 6. Discussion

### 6.1. Comparison with the literature

Our results are consistent with European actuarial literature. Cairns et al. (2009) report Lee-Carter RMSE between 0.3 and 0.6 years over 20-year horizons in Europe; our result of 0.463 years falls within this range. CBD superiority over Lee-Carter for old ages is documented by Blake et al. (2008) for UK data.

The female LC1 coverage of 82.6% falls below the theoretical 95% target, consistent with Brouhns et al. (2002), who show that treating $b_x$ as fixed in bootstrap underestimates parametric uncertainty.

### 6.2. Limitations

1. **French data only.** The methodology is transferable to other contexts (notably Morocco via ACAPS or HCP/WHO estimates) but would require comparable data granularity.
2. **Exogenous shocks not modelled.** COVID-19 (2020) falls outside confidence intervals for all models, as expected for an unforeseeable structural shock.
3. **Male LC1 coverage insufficient.** Despite the LC2 extension (+21.8 points), coverage remains below 80% for males. CBD remains the only model satisfactory for both sexes.
4. **H/F correlation.** Models are calibrated independently by sex. A multi-population approach (Li & Lee, 2005) modelling male-female mortality correlation would be a natural extension.

### 6.3. Future work

**P1. Cause-of-death decomposition.** Decomposing $e_0$ improvement into cardiovascular, cancer and external cause contributions.

**P2. Multi-population model.** Calibrating a joint model for France and emerging markets (notably Morocco) for international portfolio pricing.

**P3. Hierarchical Bayesian model.** Extending the Kalman filter to a hierarchical structure (PyMC) for more complete uncertainty propagation.

## 7. Conclusion

This paper presents a rigorous comparison of five stochastic mortality modelling approaches on French male and female data 1950–2024, with three original contributions.

First, the rolling one-step-ahead backtest over 20 years demonstrates that CBD is the best-performing model for both sexes (RMSE 0.188 years for females, 0.228 years for males; coverage 91.3% and 95.7%). Single-factor Lee-Carter is insufficient for males (47.8% coverage), while the LC2 extension improves this by +21.8 percentage points.

Second, the remarkable convergence of three models on $\ddot{a}_x$ (spread $< 0.4\%$ for females) strengthens confidence in annuity pricing, with a central premium of €5,558–5,565/year for a €100,000 capital (female 65, $i = 2\%$, 2025 horizon).

Third, for a portfolio of 100 annuities representing €14.37 M in committed capital, the total recommended provision is €595,313 (4.1% of capital), combining a stochastic provision (P99.5 CBD) of €71,298 and a Solvency II stress provision of €524,015. These figures provide a directly operational quantitative basis for asset-liability management and regulatory compliance.

## References

Blake, D., Cairns, A.J.G., & Dowd, K. (2006). Living with mortality: Longevity bonds and other mortality-linked securities. *British Actuarial Journal*, 12(1), 153–228.

Blake, D., Cairns, A.J.G., & Dowd, K. (2008). The birth of the life market. *Asia-Pacific Journal of Risk and Insurance*, 3(1), 6–36.

Brouhns, N., Denuit, M., & Vermunt, J.K. (2002). A Poisson log-bilinear regression approach to the construction of projected lifetables. *Insurance: Mathematics and Economics*, 31(3), 373–393.

Cairns, A.J.G., Blake, D., & Dowd, K. (2006). A two-factor model for stochastic mortality with parameter uncertainty: Theory and calibration. *Journal of Risk and Insurance*, 73(4), 687–718.

Cairns, A.J.G., Blake, D., Dowd, K., Coughlan, G.D., Epstein, D., Ong, A., & Balevich, I. (2009). A quantitative comparison of stochastic mortality models using data from England and Wales and the United States. *North American Actuarial Journal*, 13(1), 1–35.

Human Mortality Database (2024). University of California, Berkeley (USA), and Max Planck Institute for Demographic Research (Germany). www.mortality.org

Haberman, S., & Renshaw, A.E. (2011). A comparative study of parametric mortality projection models. *Insurance: Mathematics and Economics*, 48(1), 35–55.

Kalman, R.E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1), 35–45.

Lee, R.D., & Carter, L.R. (1992). Modeling and forecasting U.S. mortality. *Journal of the American Statistical Association*, 87(419), 659–671.

Li, N., & Lee, R. (2005). Coherent mortality forecasts for a group of populations: An extension of the Lee-Carter method. *Demography*, 42(3), 575–594.

Renshaw, A.E., & Haberman, S. (2006). A cohort-based extension to the Lee-Carter model for mortality reduction factors. *Insurance: Mathematics and Economics*, 38(3), 556–570.